

Historical Instruments and Exclusion Restrictions: A Replication Study of Roche (2020)

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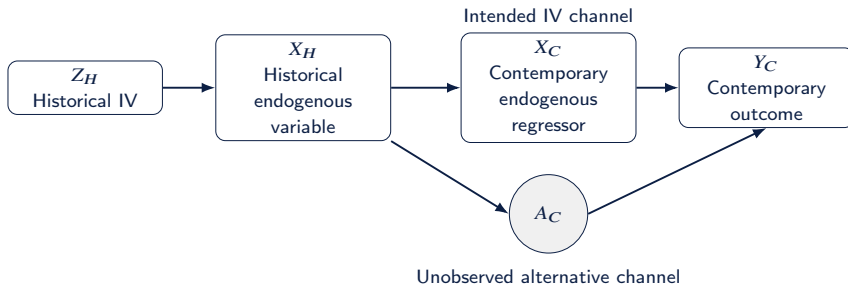
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- Instrumental Variables are widely used in empirical economics to address endogeneity issues
- In particular, **historical instruments** are often used to identify causal effects of contemporary endogenous regressors
- However, the validity of historical instruments can be questionable due to potential violations of the exclusion restriction

Why Are Historical Instruments Potentially Problematic?

- Casey and Klemp (2021) study IV regressions where historical instruments are used for contemporary endogenous regressors
- Their main point is that the exclusion restriction can fail because of the time gap between the instrument's initial impact and the measurement of the endogenous regressor



Motivating Application: Roche (2020)

- Roche (2020) asks whether greater neighborhood street connectivity increases innovation
- To address the endogeneity of street connectivity, she uses two historical housing shares as instruments:
 - ▶ housing built before 1940
 - ▶ housing built during the 1940s
- The IV estimates imply a **positive effect of street connectivity on innovation**
- Roche (2020) therefore provides a useful empirical setting for studying how qualitative conclusions depend on the exclusion restrictions imposed on historical instruments

But how much does this conclusion depend on the exact exclusion of both historical instruments?

How robust is Roche (2020)'s positive effect of street connectivity on innovation to relaxations of the exclusion restrictions on its historical instruments?

- More broadly, when the identifying assumptions of an IV design are relaxed, which relaxations preserve the qualitative conclusion?

- Conduct a wide replication of Roche (2020) by relaxing the exclusion restrictions on its historical instruments
- Apply Masten and Poirier (2021) to relax the exclusion restrictions on the two historical instruments separately and jointly
- Go beyond model falsification by asking which minimally relaxed assumptions preserve Roche's qualitative conclusion
- Characterize a conclusion-preserving region along the falsification frontier to assess where Roche's qualitative conclusion survives

Preview of Findings

- The baseline model imposing exact exclusion on both historical instruments is falsified
- Once we minimally relax those exclusion restrictions, the causal effect is only partially identified by

$$[-0.0096, 0.0504],$$

so Roche's positive sign is not preserved under every minimal relaxations

- However, the positive conclusion survives over a large portion of the falsification frontier

Part I: Theoretical Framework

- Falsifiable models: There exists a population distribution of the observable data which is inconsistent with the model
- What should researchers do when their baseline model is falsified?
- Masten and Poirier (2021) provides a way to salvage a falsified baseline model by considering continuous relaxations of the baseline assumptions
- By sufficiently weakening the assumptions, a falsified baseline model becomes non-falsified

- Suppose the model has L assumptions
- For each assumption $l \in \{1, 2, \dots, L\}$, define a class of assumptions indexed by δ_l ($\delta_l = 0$ means that baseline assumption l is imposed)
- Using the definition above, $\delta'_l \geq \delta''_l$ means that δ'_l is weaker than δ''_l
- Let $\delta = (\delta_1, \dots, \delta_L)$ and $\delta = 0$ mean the baseline model

The falsification frontier is the set

$$FF \equiv \{\delta \in \mathcal{D} : \delta \in \mathcal{D}_{nf} \text{ and for any } \delta' < \delta, \delta' \in \mathcal{D}_f\}$$

- where $\mathcal{D}_f$ is set of all assumptions which are falsified
- $\mathcal{D}_{nf}$ is set of all assumptions which are non-falsified
- For any $\delta_0 \in FF$, the model with assumption δ' is falsified if $\delta' < \delta_0$
- The frontier represents **the smallest relaxation necessary to reconcile the model with the data**

- Let $\Theta_I(\delta)$ denote the identified set for a parameter of interest, given the model with assumption δ
- Then we can define falsification adaptive set (FAS) by

$$\bigcup_{\delta \in FF} \Theta_I(\delta)$$

- FAS summarizes every parameter value supported by minimally relaxed assumptions

Linear Models with Multiple IVs

For each $(d, z) \in \text{Supp}(D, Z)$, consider the following linear model:

$$Y(d, z) = d'\theta + z'\gamma + U$$

- where $d \in \mathbb{R}^K$ and $z \in \mathbb{R}^L$
- Assumption 1: $\text{rank}(\text{Cov}(Z, d)) = K$
- Assumption 2: $\text{Var}(Z)$ is invertible
- Assumption 3: $\text{Cov}(Z, U) = 0$
- Assumption 4: $\gamma = 0$
- Assumption 4' (Partial Exclusion): There are known constants $\delta_l \geq 0$ such that $|\gamma_l| \leq \delta_l \quad \forall l = 1, 2, \dots, L$

Theorem

(Identified Set for θ)

$$\Theta_I(\delta) = \{b \in \mathbb{R}^K : -\delta \leq \text{Var}(Z)^{-1}(\text{Cov}(Z, Y) - \text{Cov}(Z, D)b) \leq \delta\}$$

- Let $\psi \equiv \text{Var}(Z)^{-1}\text{Cov}(Z, Y)$ and $\pi \equiv \text{Var}(Z)^{-1}\text{Cov}(Z, D)$
- The identified set when $K = 1$ can be rewritten in terms of the components of ψ and π (cf. Corollary 1)
- The model is falsified if $B(\delta) = \emptyset$

Proposition

(Falsification Frontier) Under Assumption 1-3,

$$FF = \{\delta \in \mathbb{R}_+^L : \delta_l = |\psi_l - b\pi_l|, l = 1, 2, \dots, L, b \in [\min_{l:\pi_l \neq 0} \frac{\psi_l}{\pi_l}, \max_{l:\pi_l \neq 0} \frac{\psi_l}{\pi_l}]\}$$

Theorem

(Falsification Adaptive Set) Under Assumption 1-3,

$$FAS = [\min_{l:\pi_l \neq 0} \frac{\psi_l}{\pi_l}, \max_{l:\pi_l \neq 0} \frac{\psi_l}{\pi_l}]$$

- Lemma 1 shows that $\frac{\psi_l}{\pi_l}$ is equivalent to 2SLS coefficient on D using Z_l as the excluded IV and Z_{-l} as controls

Part II: Empirical Application

Roche (2020): Setting

- Research Question: Does greater neighborhood street connectivity increase innovation?
- Unit of Analysis: U.S. census block group
- Outcome: Log number of granted patents applied for between 2011 and 2013
- Road Connectivity: Total miles of multimodal streets in a block group divided by its area
- Main idea: more connected street networks may facilitate interpersonal interactions and knowledge spillovers, thereby increasing innovation

Roche (2020) estimates

$$Y_i = \alpha + \theta D_i + X_i' \beta + U_i,$$

where

- Y_i : log number of patents in block group i
- D_i : road connectivity
- X_i : social activity, formal knowledge, human capital, sociodemographic, and physical geography controls
- U_i : unobserved determinants of innovation
- **Parameter of interest:** θ , the causal effect of road connectivity on innovation

Endogeneity Issues and IV Approach

- Road connectivity is unlikely to be exogenous:
 - ▶ developed areas may have both **denser street networks** and **greater innovation potential**
 - ▶ local amenities or favorable initial conditions may attract skilled workers and innovative activity
- Roche (2020) uses two historical housing variables as instruments:
 - ▶ Z_{1i} : the share of housing built before 1940 in block group i (HUp_{pre1940})
 - ▶ Z_{2i} : the share of housing built during 1940 in block group i (HU_{1940s})
- The original IV design requires both historical instruments to affect innovation only through road connectivity

Concerns about Exclusion Restrictions

Historical Housing $\longrightarrow$ Road Connectivity $\longrightarrow$ Innovation

Historical Housing $\rightarrow$ Persistent Urban Development $\longrightarrow$ Agglomeration $\longrightarrow$ Innovation

- Historically developed neighborhoods may remain relatively dense because urban development is persistent
- Urban density itself may increase innovation through agglomeration and knowledge spillovers
- Historical housing shares may therefore affect current innovation through channels other than road connectivity

Relaxing the Exclusion Restrictions

After partialling out the controls X , consider

$$Y(d, z) = d\theta + z_1\gamma_1 + z_2\gamma_2 + U.$$

Exact exclusion imposes

$$\gamma_1 = \gamma_2 = 0.$$

Relaxing the Exclusion Restrictions

Instead, following Masten and Poirier (2021), allow

$$|\gamma_1| \leq \delta_1, \quad |\gamma_2| \leq \delta_2.$$

- δ_1 : allowed direct effect of HUpre1940
- δ_2 : allowed direct effect of HU1940s
- $(\delta_1, \delta_2) = (0, 0)$: original Roche (2020) exclusion restrictions
- Increasing δ_ℓ weakens the exclusion restriction for instrument ℓ

Falsification Frontier

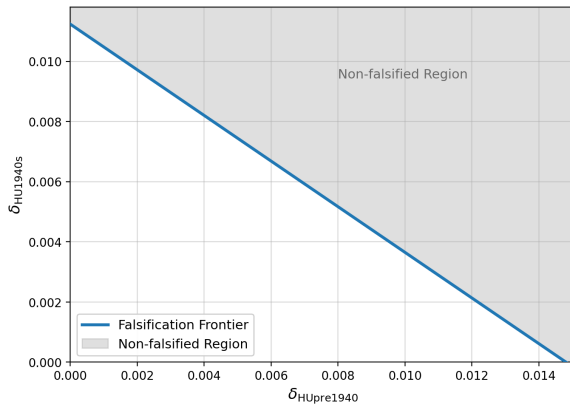


Figure 1: Falsification Frontier

Falsification Adaptive Set (FAS)

Panel A. Estimates	
Baseline 2SLS	0.0426** (0.0188)
HUpred1940 as IV, control HU1940s	0.0504*** (0.0193)
HU1940s as IV, control HUpred1940	-0.0096 (0.0449)
Panel B. Falsification adaptive set	
Relevant instruments	HUpred1940, HU1940s
First-stage F -statistics	55.32, 14.78
FAS	$[-0.0096, 0.0504]$

Why Do the Two FAS Endpoints Differ?

- The two controlled just-identified specifications associated with different maintained exclusions deliver opposite signs
- The HU1940s-based specification has substantially weaker first-stage relevance:

$$F = 14.78 \quad \text{vs.} \quad 55.32$$

- Boeing (2021) identifies around 1940 as a **rupture point** in U.S. street-network development
- Pre-1940 housing therefore captures Roche's road-connectivity mechanism more directly
- **This explains the relevance asymmetry, not IV validity**

Maintained Exclusion Restriction Assumption

- U.S. cities experienced substantial demographic and urban restructuring around and after World War II
- HU1940s may therefore capture wartime industrial development and subsequent urban expansion
- Boeing (2021) also documents major changes in urban planning and automobile-oriented infrastructure beginning around 1940
- These historical forces could affect later innovation through channels other than road connectivity

Interpretation

The data do not establish that HUpre1940 satisfies the exclusion restriction while HU1940s does not. Rather, the historical evidence gives us a reason to place greater interpretive weight on specifications that maintain HUpre1940 as the excluded instrument.

Conclusion-Preserving Region

- The FAS tells us which parameter values survive minimal relaxations, but not where a particular qualitative conclusion survives
- The endpoint estimates show that recovery of Roche's positive sign depends on the maintained exclusion restriction
- Which minimally non-falsified relaxations preserve Roche's qualitative conclusion?

Definition

(Conclusion-Preserving Region) For a given δ , let $\Theta(\delta)$ denote the identified set for θ . The conclusion-preserving region for $\inf \Theta(\delta) \geq 0$ is $FF^+ = \{\delta \in FF : \inf \Theta(\delta) \geq 0\}$.

Conclusion-Preserving Region

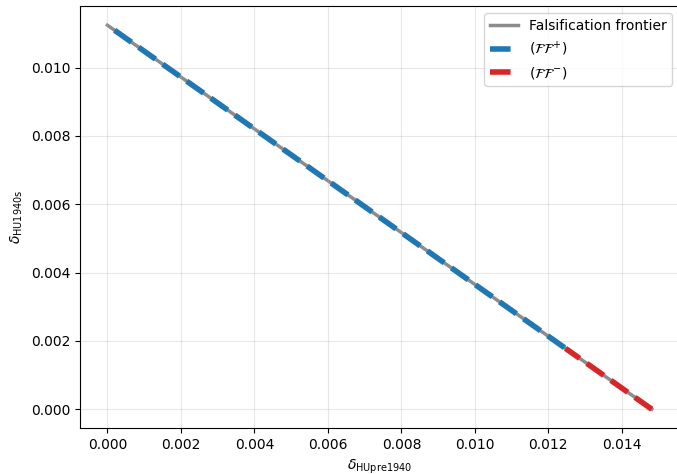


Figure 2: Conclusion-Preserving Region

- The baseline model imposing exact exclusion on both historical instruments is falsified
- Nevertheless, the positive conclusion is preserved over a large portion of the falsification frontier
- Conditional on maintaining HUp_{pre1940} as the excluded instrument, the estimated effect remains positive and similar in magnitude to Roche (2020)
- We show that sensitivity analysis shifts the question from whether the instruments are valid to which conclusions survive minimally non-falsified relaxations

Thank You